

Algorithms 2026: Quiz 1

Answer both questions. Total Marks: $2 \times 5 = 10$

1. Let S be an unsorted set of n distinct integers. Given any $1 \leq r \leq n$, we know how to find in $O(n)$ time the number that occurs in r -th position when elements of S are listed in sorted order. Suppose we are given $k \geq 2$ different ranks $r_1, \dots, r_k$. Show an $O(n \log k)$ algorithm to find the k different numbers that occur in $r_1, r_2, \dots, r_k$ positions when elements of S are listed in sorted order.
2. Let G be an undirected graph on n vertices, all edge have weight 1. Show that the number of distinct global minimum cuts in G is at most $\binom{n}{2}$. Is this a tight bound? Why/why not?